

Minimal bad SCC setup for the general PSC

Abstract

We formalize the standing counterexample object for the general PSC: a minimal reachable noncoincident strongly connected component (SCC) in the small-regime balanced-pair automaton. All structural properties inherited from the completed program components are collected. This note is the canonical setup reference for all subsequent SCC-level arguments.

1 Standing hypotheses

Standing Setup (Minimal bad SCC). The following is assumed throughout.

- (S1) σ is a primitive irreducible Pisot substitution on $\mathcal{A}$ with $|\mathcal{A}| \geq 4$, $\det(M) \neq 0$, and $\mathbb{Z}$ -independent tile lengths.
- (S2) $\text{SRE}(|\mathcal{A}|)$ fails: there exists a periodic noncoincident cycle in the small regime of $\mathcal{B}_\sigma$.
- (S3) $\mathcal{C}$ is a *minimal* reachable noncoincident SCC in the small-regime subautomaton $\mathcal{B}_\sigma^{\text{sm}}$. Minimality means: no proper sub-SCC of $\mathcal{C}$ is itself a noncoincident SCC.
- (S4) $|\mathcal{A}|$ is the *smallest alphabet size* for which SRE fails (minimal-alphabet counterexample).

2 Inherited structural properties

The completed program components give the following properties of $\mathcal{C}$ for free.

Proposition 2.1 (Permutation pairs). *Every state $(u, v) \in \mathcal{C}$ satisfies $\text{Parikh}(u) = \text{Parikh}(v)$ and $u \neq v$. In particular, u is a permutation of v .*

Proof. $\mathbb{Z}$ -independence of tile lengths. □

Proposition 2.2 (Full support). *Every state $(u, v) \in \mathcal{C}$ has $\text{supp}(u) \cup \text{supp}(v) = \mathcal{A}$.*

Proof. If $\text{supp}(u) \cup \text{supp}(v) \subsetneq \mathcal{A}$, the restricted ambiguity inheritance theorem propagates the noncoincidence to a smaller alphabet, contradicting (S4). □

Proposition 2.3 (Bounded word length). *There exists $B = B(\sigma)$ such that $|u| = |v| \leq B$ for every $(u, v) \in \mathcal{C}$.*

Proof. The finiteness engine gives a uniform small-regime bound. □

Proposition 2.4 (Recurrence). *Every state in $\mathcal{C}$ returns to itself: for each $(u, v) \in \mathcal{C}$, there exists $k \geq 1$ such that $\Phi_\sigma^k(u, v) = (u, v)$.*

Proof. $\mathcal{C}$ is a finite SCC, so every state is on a cycle. $\square$

Proposition 2.5 (Reachability). *Every state in $\mathcal{C}$ is reachable from the initial state of $\mathcal{B}_\sigma$ (the root balanced pair).*

Proof. By definition of the reachable subautomaton. $\square$

Proposition 2.6 (Inversion dynamics). *For every $(u, v) \in \mathcal{C}$ with child $(u', v') = \Phi_\sigma(u, v)$,*

$$\Pi_{\text{inv}}(\sigma(u), \sigma(v)) = (M \otimes M) \Pi_{\text{inv}}(u, v),$$

and $\Pi_{\text{inv}}(u', v')$ is the restriction of the inflated inversion profile to the child-retained coordinates.

Proof. Inversion inflation formula (Theorem 3.2 of the dichotomy note). $\square$

3 The common-cut bracket structure on $\mathcal{C}$

For each state $(u, v) \in \mathcal{C}$, the inflate step gives tile sequences $\sigma(u)$ and $\sigma(v)$. The common geometric cuts $C_\cap$ partition the inflated pair into common-cut intervals.

Proposition 3.1 (Common-cut intervals). *At each state $(u, v) \in \mathcal{C}$:*

- (i) *Padding intervals have identical source words on both sides (by P1 and $\mathbb{Z}$ -independence).*
- (ii) *Child-overlapping intervals have source words $\mathbf{a}_j \neq \mathbf{b}_j$ with $\text{Parikh}(\mathbf{a}_j) = \text{Parikh}(\mathbf{b}_j)$.*
- (iii) *The child (u', v') is extracted from within the child-overlapping region.*
- (iv) *The child-overlapping intervals have bounded total source-word length $\leq B$.*

Definition 3.2 (Bracket type). The *bracket type* of a state $(u, v) \in \mathcal{C}$ is the finite data:

- the number and types of common-cut intervals;
- the Parikh vectors $\text{Parikh}(\mathbf{a}_j)$ on each interval;
- which intervals are padding vs. child-overlapping;
- the boundary germs (source letters adjacent to child regions).

Since $|u| \leq B$, there are finitely many bracket types.

Proposition 3.3 (Recurrent bracket types). *In a minimal bad SCC, every bracket type that appears at some state appears infinitely often along every cycle through that state.*

Proof. SCC recurrence: every state returns, so every local configuration repeats. $\square$

4 The SCC return map

Definition 4.1 (SCC return map). For a state $s = (u, v) \in \mathcal{C}$, the k -step return map is Φ_σ^k where k is the minimal period of s in $\mathcal{C}$. The return substitution is σ^k .

Proposition 4.2 (Return substitution properties). *The return substitution σ^k is again primitive irreducible Pisot with:*

- Perron eigenvalue $\lambda^k > 1$;
- substitution matrix M^k ;
- tile lengths ℓ (unchanged);
- recognizability radius $R(\sigma^k) = R(\sigma)$ (Mossé recognizability is inherited by powers).

Remark 4.3 (Inflation under the return map). Under the return substitution σ^k , the padding grows by λ^k while the recognizability radius stays fixed. So for large enough k , the boundary padding of every common-cut bracket exceeds $R(\sigma)$. This is the “recognizability upgrade” from the bracket lemma, now automatic from the SCC return structure.

5 The inversion dynamics on $\mathcal{C}$

Proposition 5.1 (Return-map inversion formula). *For a state $s = (u, v) \in \mathcal{C}$ with k -step return,*

$$\Pi_{\text{inv}}(\sigma^k(u), \sigma^k(v)) = (M \otimes M)^k \Pi_{\text{inv}}(u, v).$$

The child after k steps has inversion profile $\Pi_{\text{inv}}(u, v)$ again (periodicity). So the return map on inversion space satisfies:

$$\Pi_{\text{inv}}(u, v) = R_k((M \otimes M)^k \Pi_{\text{inv}}(u, v)),$$

where R_k is the k -step child-restriction operator.

Corollary 5.2 (Fixed-point equation). $\Pi_{\text{inv}}(u, v) \neq 0$ is a nonzero solution of

$$x = R_k((M \otimes M)^k x).$$

Since $(M \otimes M)^k$ has spectral radius ρ^k on the off-diagonal space, and the child restriction R_k is a (nonlinear) sub-selection, this fixed-point equation constrains x to a very specific locus.

6 The candidate attack routes at SCC level

- (RS1) **Common-cut recurrence.** The bracket type recurs at every return. If the common-cut structure degrades in any measurable way along the cycle (more padding, fewer disagreement intervals, smaller child Parikh), the cycle cannot close. This is the Front A route upgraded to SCC level.
- (RS2) **Monotone invariant.** Find $I : \mathcal{C} \rightarrow \Xi$ (finite poset) such that I strictly decreases along edges unless coincidence occurs. Recurrence then forces coincidence. This is the cleanest abstract route.

- (RS3) **Inversion fixed-point impossibility.** Show that the fixed-point equation $x = R_k((M \otimes M)^k x)$ has no nonzero full-support solution for any child-restriction operator R_k arising from a reachable balanced pair. This is the Front B route (ERI) at SCC level.
- (RS4) **Support-drop under return.** Show that after one full return through the SCC, some letter pair drops from the active support of Π_{inv} . After finitely many returns, the support is empty: coincidence. This combines inversion dynamics with the SCC recurrence.

7 Honest assessment

What is formalized. The standing counterexample object (minimal bad SCC) with all inherited properties: permutation pairs, full support, bounded word length, recurrence, reachability, common-cut bracket structure, inversion dynamics, return-map fixed-point equation.

What is open. Any of RS1–RS4 that would exclude the standing counterexample. The closest route is still RS1 (common-cut recurrence), but RS4 (support-drop) is the most natural use of the inversion engine at SCC level.

The one theorem that would close the general PSC.

No minimal reachable noncoincident SCC exists in the small-regime automaton of any primitive irreducible Pisot substitution.